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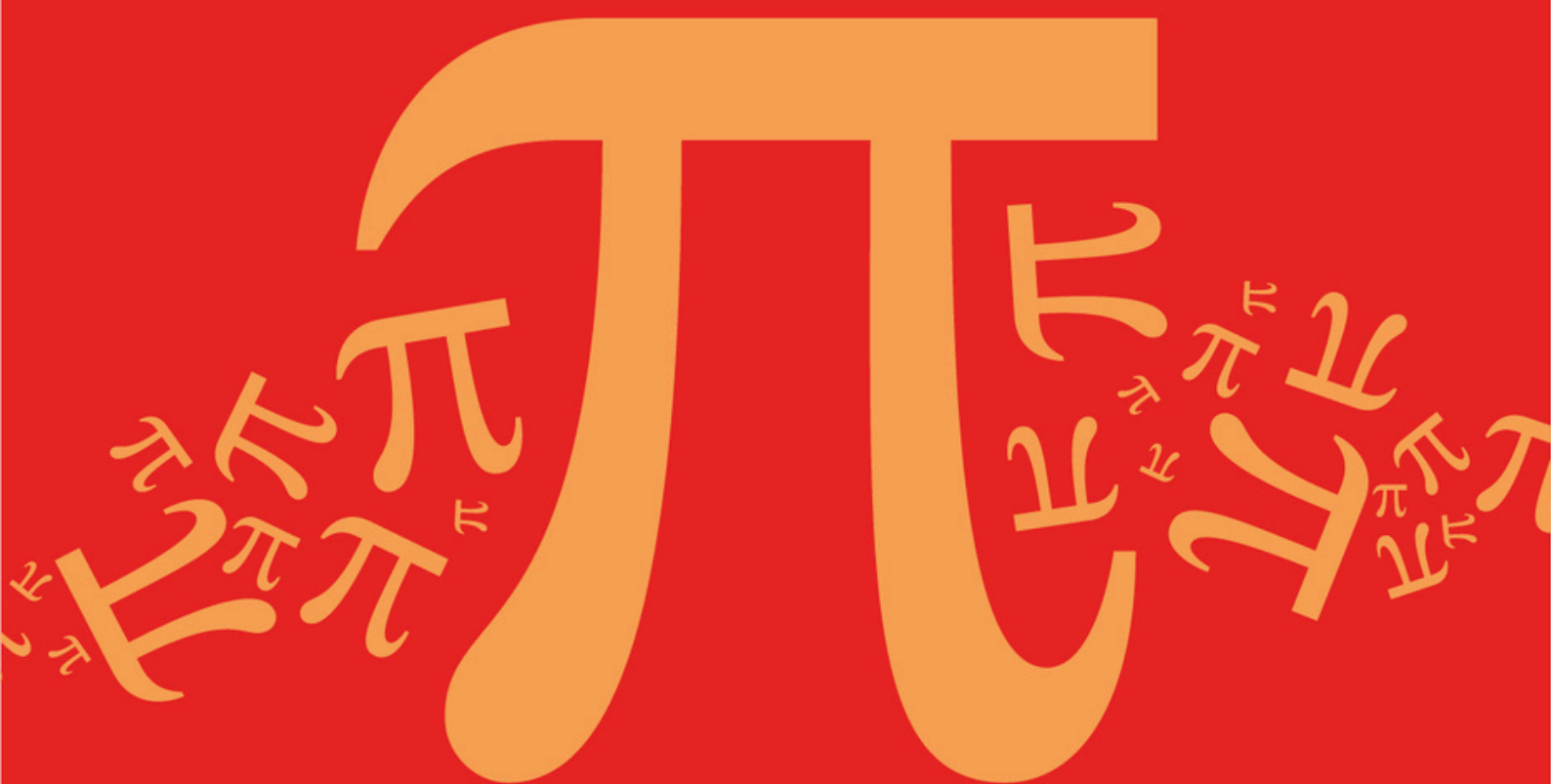
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AQA A Level Further Mathematics (7367)



Topic

DB: Networks

Sub topic

DB2: Network Optimisation/Spanning Trees

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

DB: Networks

Sub topic

DB2: Network Optimisation/Spanning Trees

EXAM PAPERS PRACTICE

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

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Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

An animal sanctuary has a rainwater collection site.

The manager of the sanctuary is installing a pipe system to connect the rainwater collection site to five other sites in the sanctuary.

Each site does **not** need to be connected directly to the rainwater collection site.

There are nine possible routes between the sites that are suitable for water pipes.

The distances, in metres, of the nine possible routes are given in the table below.

From/To	Henhouse (H)	Goatshed (G)	Kennels (K)	Cattery (C)
Rainwater collection site (R)	840	810	520	370
Cattery (C)	–	680	610	
Duckpond (D)	480	310		
Goatshed (G)	150			

Water pipe costs 60 pence per metre.

Find the minimum cost of connecting all the sites to the rainwater collection site.

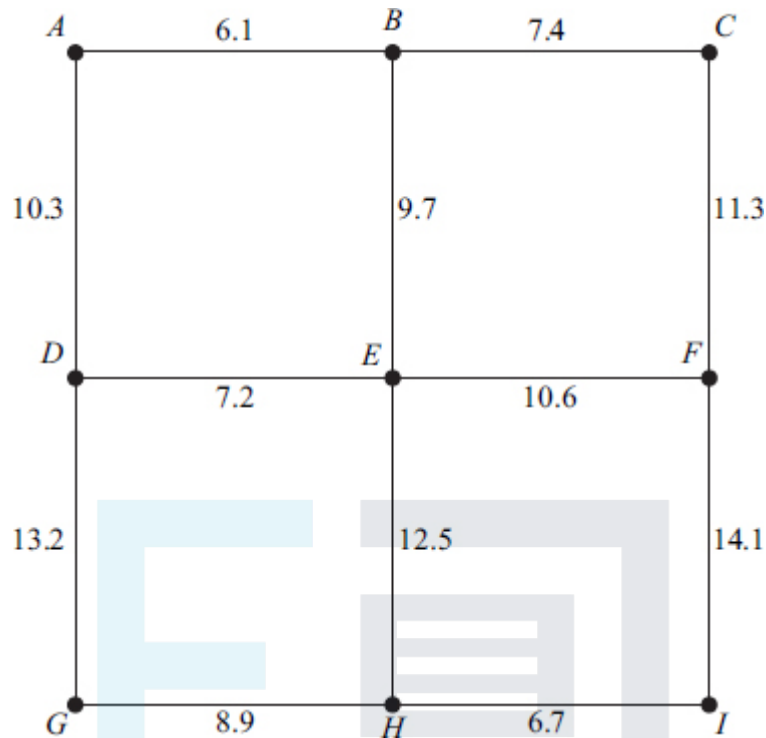
Fully justify your answer.

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Q2.

The following network shows the lengths, in miles, of roads connecting nine villages, A, B, ..., I.

A programme of resurfacing some roads is undertaken to ensure that each village can access all other villages along a resurfaced road, while keeping the amount of road to be resurfaced to a minimum.



- (a) (i) Use Prim's algorithm starting from A , showing the order in which you select the edges, to find a minimum spanning tree for the network.
- (ii) State the length of your minimum spanning tree.
- (iii) Draw your minimum spanning tree.

(7)

- (b) Given that Prim's algorithm is used with different start vertices, state the final edge to be added to the minimum spanning tree if:

- (i) the start vertex is E ;
- (ii) the start vertex is G .

(2)

- (c) Given that Kruskal's algorithm is used to find the minimum spanning tree, state which edge would be:

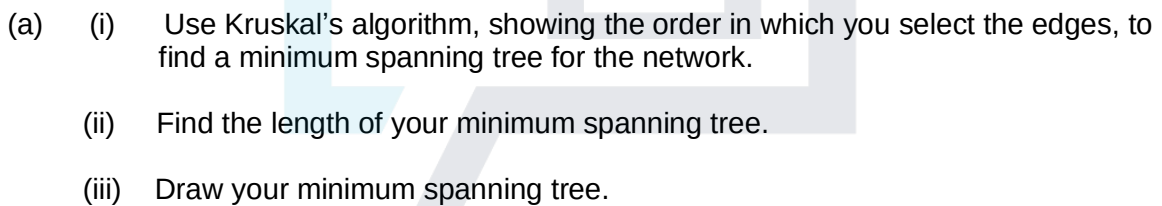
- (i) the first to be included in the tree;
- (ii) the last to be included in the tree.

(2)

(Total 11 marks)

Q3.

The following network shows the lengths, in miles, of roads connecting ten villages, $A, B, C, \dots, J$.

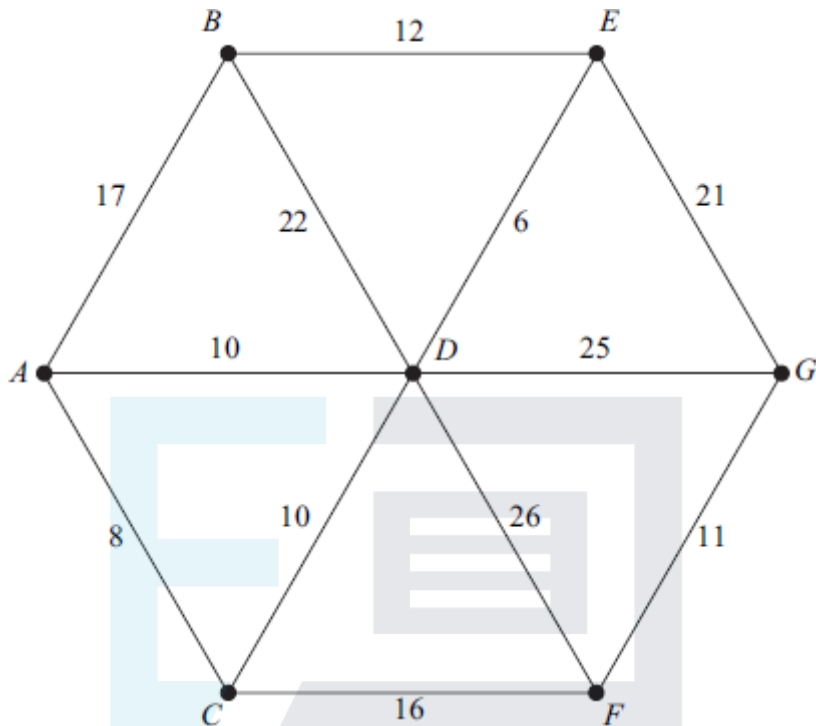


(b) Prim's algorithm from different starting points produces the same minimum spanning tree. State the final edge that would be added to complete the minimum spanning tree if the starting point were:

- (2)**

(Total 9 marks)

The following network shows the roads connecting seven villages, $A, B, C, \dots, G$. The number on each edge represents the length, in miles, between a pair of villages.



- (a) Use Kruskal's algorithm to find a minimum spanning tree for the network.

(5)

- (b) State the length of your minimum spanning tree.

(1)

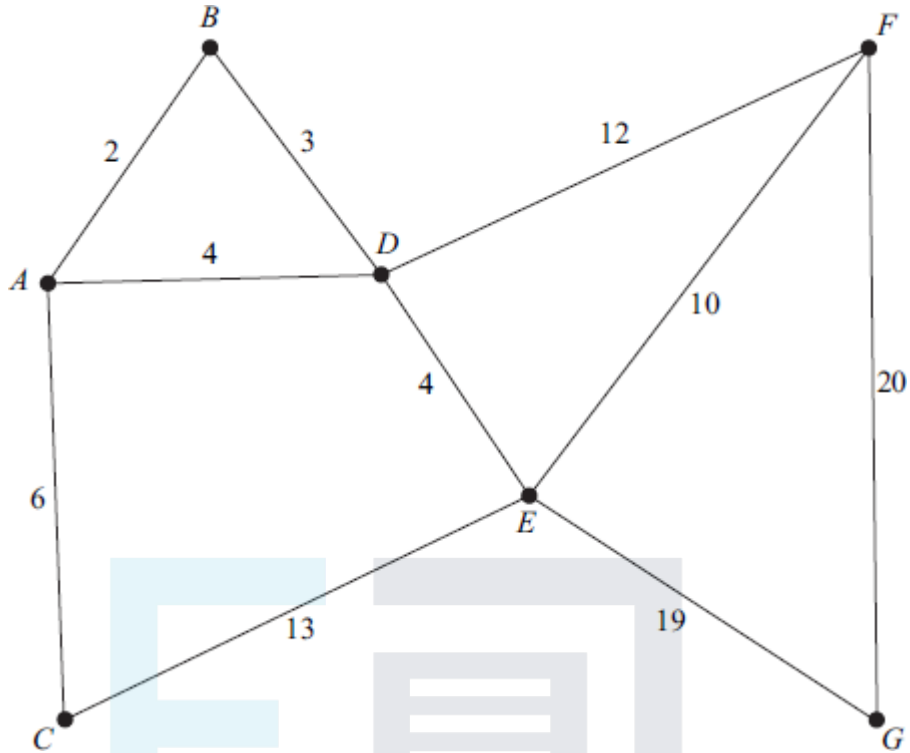
- (c) There are two minimum spanning trees for this network. Draw both of these minimum spanning trees.

(3)

(Total 9 marks)

Q5.

The diagram shows the locations of some schools. The number on each edge shows the distance, in miles, between pairs of schools.



Sam, an adviser, intends to travel from one school to the next until he has visited all of the schools, before returning to his starting school. The shortest distances for Sam to travel between some of the schools are shown in **Table 1** below.

(a) Complete **Table 1**.

Table 1

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>
<i>A</i>	–	2	6	4		16	27
<i>B</i>	2	–	8	3		15	26
<i>C</i>	6	8	–	10		22	32
<i>D</i>	4	3	10	–		12	23
<i>E</i>					–		
<i>F</i>	16	15	22	12		–	20
<i>G</i>	27	26	32	23		20	–

(2)

(b) (i) On the completed **Table 1**, use the nearest neighbour algorithm, starting from *B*, to find an upper bound for the length of Sam's tour.

(4)

(ii) Write down Sam's actual route if he were to follow the tour corresponding to the answer in part (b)(i).

(2)

- (iii) Using the nearest neighbour algorithm, starting from each of the other vertices in turn, the following upper bounds for the length of Sam's tour were obtained: 77, 77, 77, 76, 77 and 76.

Write down the best upper bound.

(1)

- (c) (i) On **Table 2** below, showing the order in which you select the edges, use Prim's algorithm, starting from *A*, to find a minimum spanning tree for the schools *A*, *B*, *C*, *D*, *F* and *G*.

Table 2

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>F</i>	<i>G</i>
<i>A</i>	–	2	6	4	16	27
<i>B</i>	2	–	8	3	15	26
<i>C</i>	6	8	–	10	22	32
<i>D</i>	4	3	10	–	12	23
<i>F</i>	16	15	22	12	–	20
<i>G</i>	27	26	32	23	20	–

(4)

- (ii) Hence find a lower bound for the length of Sam's minimum tour.

(3)

- (iii) By deleting each of the other vertices in turn, the following lower bounds for the length of a minimum tour were found: 50, 48, 52, 51, 56 and 64.

Write down the best lower bound.

(1)

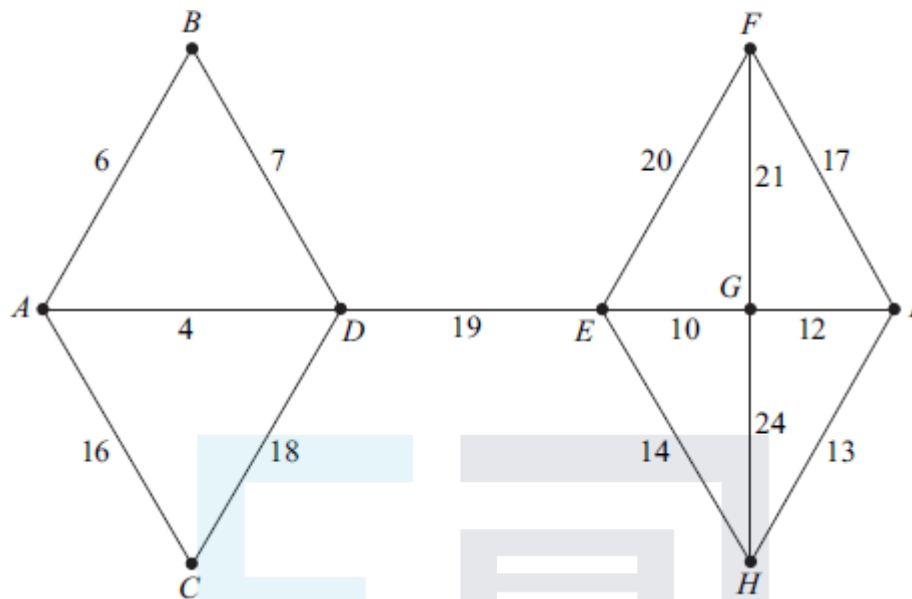
- (d) Given that the length of a minimum tour is T miles, use your answers to parts (b) and (c) to write down the smallest interval within which you know T must lie.

(2)

(Total 19 marks)

Q6.

The following network shows the lengths, in miles, of roads connecting nine villages, *A*, *B*, ..., *I*.

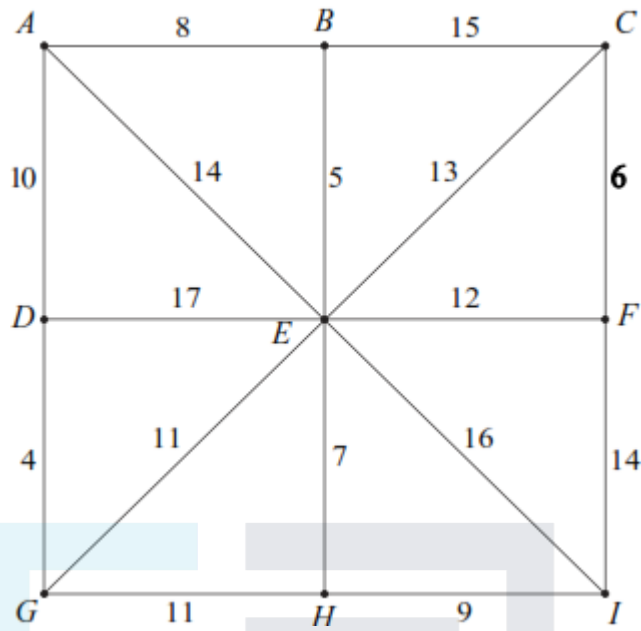


- (a) (i) Use Prim's algorithm starting from A , showing the order in which you select the edges, to find a minimum spanning tree for the network. (4)
- (ii) State the length of your minimum spanning tree. (1)
- (iii) Draw your minimum spanning tree. (2)
- (b) Prim's algorithm from different starting points produces the same minimum spanning tree for this network. State the final edge that would complete the minimum spanning tree using Prim's algorithm:
- (i) starting from D ; (1)
- (ii) starting from H . (1)

(Total 9 marks)

Q7.

The following network shows the lengths, in miles, of roads connecting nine villages, A , B , ..., I .



- (a) (i) Use Prim's algorithm starting from E , showing the order in which you select the edges, to find a minimum spanning tree for the network. (4)
- (ii) State the length of your minimum spanning tree. (1)
- (iii) Draw your minimum spanning tree. (2)
- (b) On a particular day, village B is cut off, so its connecting roads cannot be used. Find the length of a minimum spanning tree for the remaining eight villages. (2)
- (Total 9 marks)**

Q8.

A group of eight friends, A, B, C, D, E, F, G and H , keep in touch by sending text messages. The cost, in pence, of sending a message between each pair of friends is shown in the following table.

	A	B	C	D	E	F	G	H
A	–	15	10	12	16	11	14	17
B	15	–	15	14	15	16	16	15
C	10	15	–	11	10	12	14	9
D	12	14	11	–	11	12	14	12
E	16	15	10	11	–	13	15	14

<i>F</i>	11	16	12	12	13	–	14	8
<i>G</i>	14	16	14	14	15	14	–	13
<i>H</i>	17	15	9	12	14	8	13	–

One of the group wishes to pass on a piece of news to all the other friends, either by a direct text or by the message being passed on from friend to friend, at the minimum total cost.

- (a) (i) Use Prim's algorithm starting from *A*, showing the order in which you select the edges, to find a minimum spanning tree for the table. (4)
- (ii) Draw your minimum spanning tree. (2)
- (iii) Find the minimum total cost. (1)
- (b) Person *H* leaves the group. Find the new minimum total cost. (2)

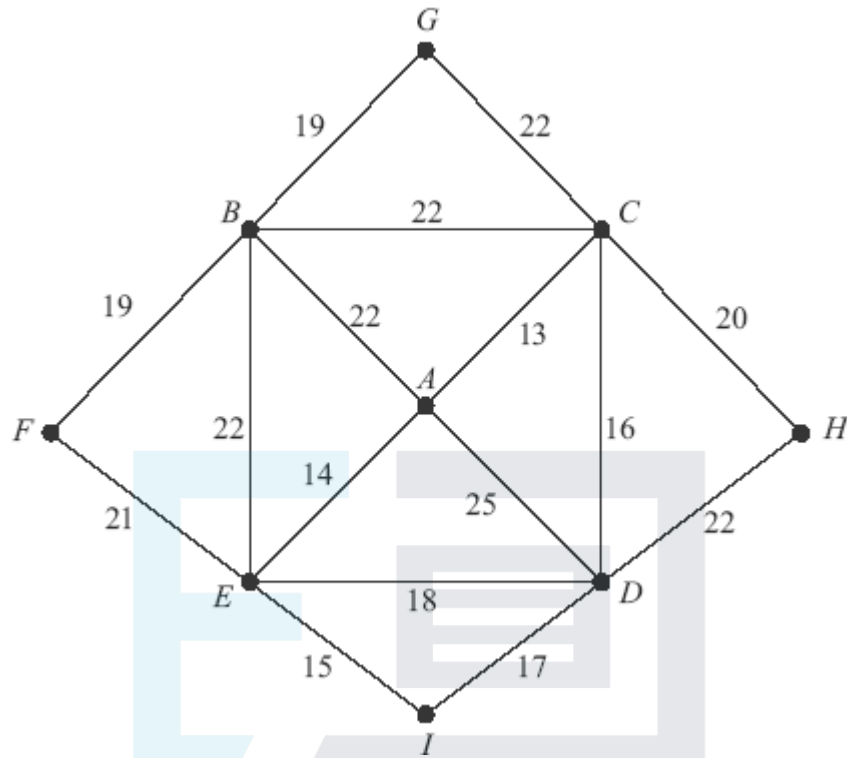
(Total 9 marks)

Q9.

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In Paris, there is a park where there are statues of famous people; there are many visitors each day to this park. Lighting is to be installed at nine places, *A*, *B*, ..., *I*, in the park. The places have to be connected either directly or indirectly by cabling, to be laid alongside the paths, as shown in the diagram.

The diagram shows the length of each path, in metres, connecting adjacent places.



Total length of paths = 307 metres

- (a) (i) Use Prim's algorithm, starting from A, to find the minimum length of cabling required.

(5)

- (ii) State this minimum length.

(1)

- (iii) Draw the minimum spanning tree.

(2)

- (b) A security guard walks along all the paths before returning to his starting place. Find the length of an optimal Chinese postman route for the guard.

(6)

(Total 14 marks)

Q10.

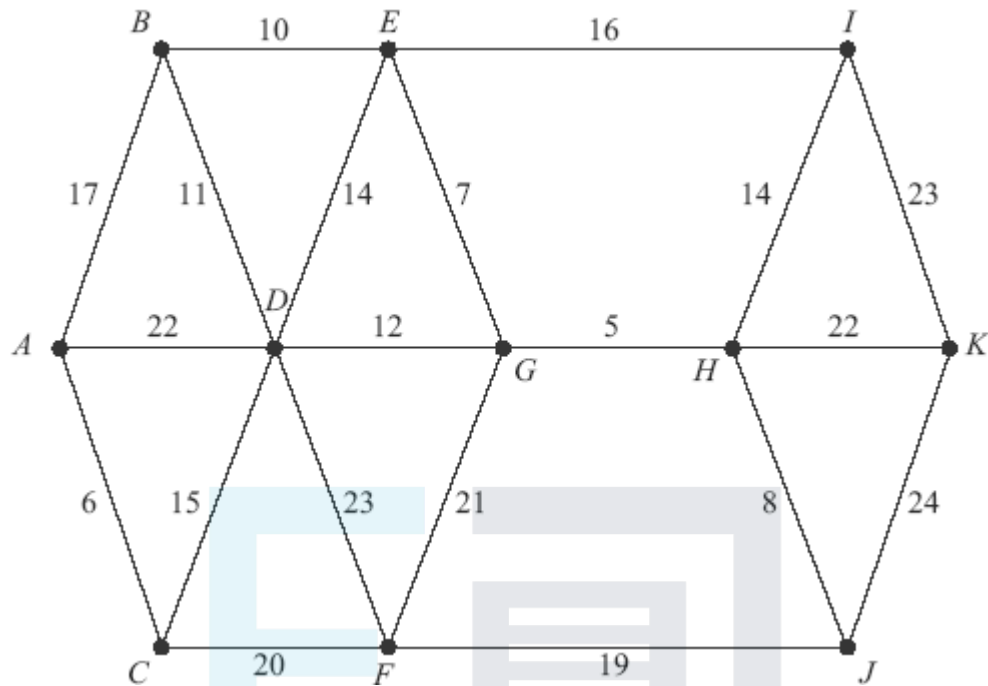
The network shows 10 towns. The times, in minutes, to travel between pairs of towns are indicated on the edges.



- (1)**

Q11.

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- (a) Starting from G and showing your working at each stage, use Prim's algorithm to find a minimum spanning tree for the network.

(6)

- (b) State the length of your minimum spanning tree.

(1)

- (c) Draw your minimum spanning tree.

(3)

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Q12.

A connected graph G has five vertices and has eight edges with lengths 8, 10, 10, 11, 13, 17, 17 and 18.

- (a) Find the minimum length of a minimum spanning tree for G .

(2)

- (b) Find the maximum length of a minimum spanning tree for G .

(2)

- (c) Draw a sketch to show a possible graph G when the length of the minimum spanning tree is 53.

(3)

(Total 7 marks)

Q13.

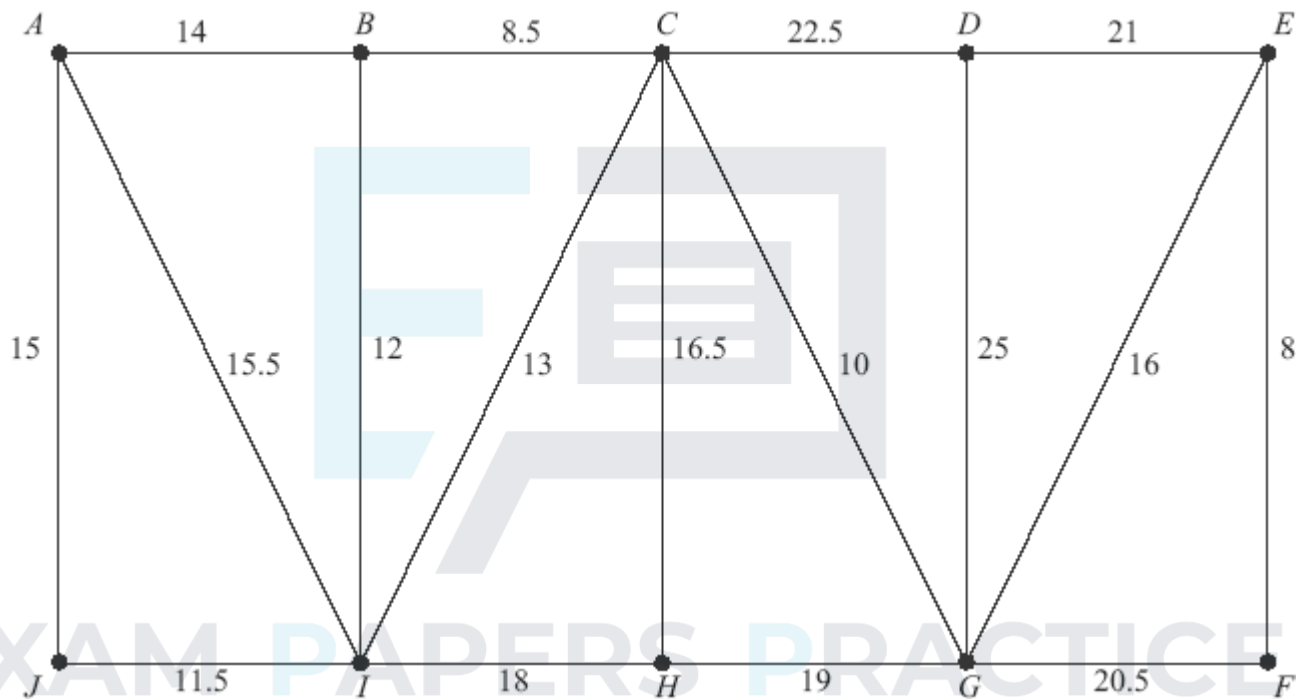
- (a) (i) State the number of edges in a minimum spanning tree for a network with 10 vertices.

(1)

- (ii) State the number of edges in a minimum spanning tree for a network with n vertices.

(1)

- (b) The following network has 10 vertices: $A, B, \dots, J$. The number on each edge represents the distance between a pair of adjacent vertices.



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- (i) Use Kruskal's algorithm to find the minimum spanning tree for the network.

(5)

- (ii) State the length of your minimum spanning tree.

(1)

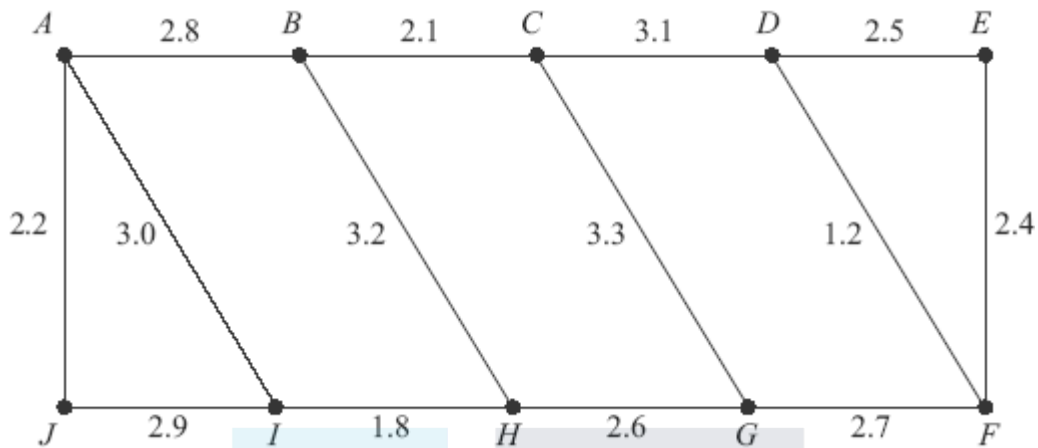
- (iii) Draw your minimum spanning tree.

(2)

(Total 10 marks)

Q14.

The diagram shows 10 bus stops, $A, B, C, \dots, J$, in Geneva. The number on each edge represents the distance, in kilometres, between adjacent bus stops.

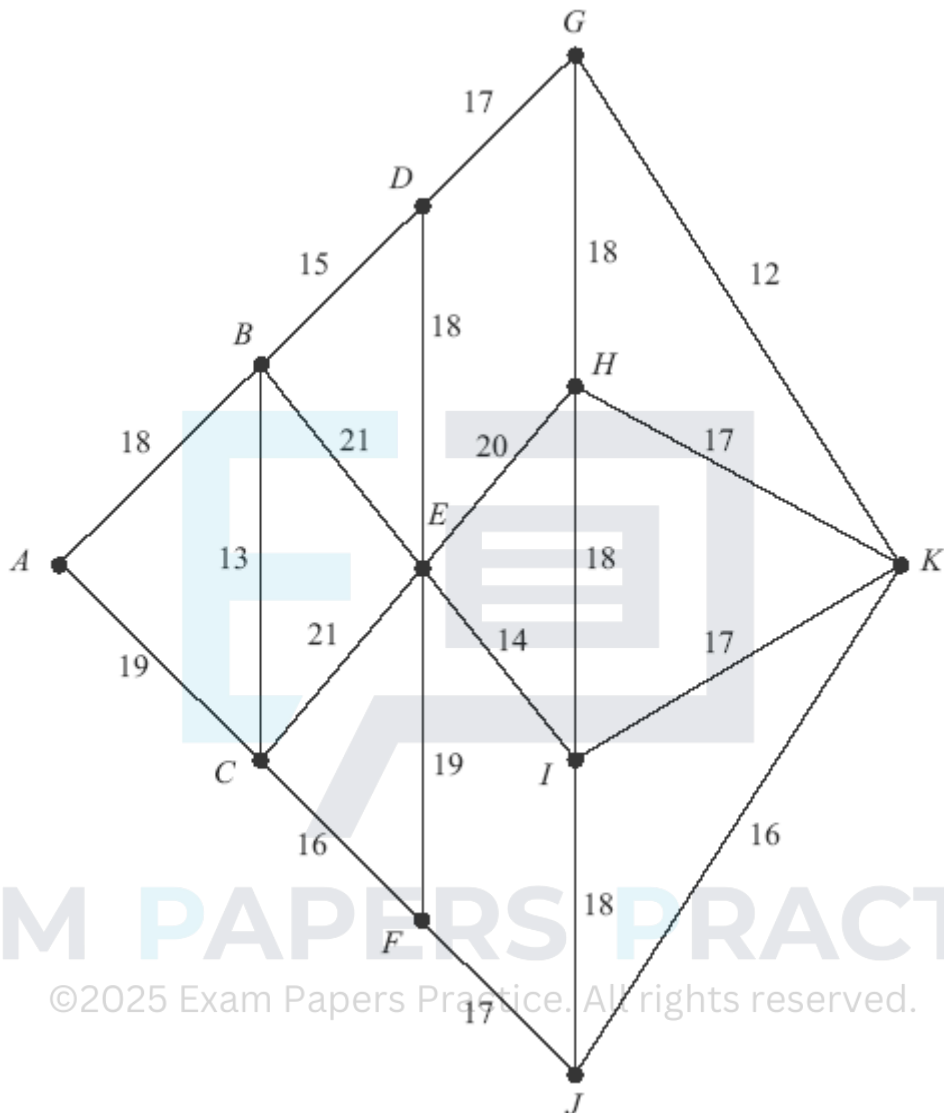


The city council is to connect these bus stops to a computer system which will display waiting times for buses at each of the 10 stops. Cabling is to be laid between some of the bus stops.

- (a) Use Kruskal's algorithm, showing the order in which you select the edges, to find a minimum spanning tree for the 10 bus stops. (5)
 - (b) State the minimum length of cabling needed. (1)
 - (c) Draw your minimum spanning tree. (2)
 - (d) If Prim's algorithm, starting from A , had been used to find the minimum spanning tree, state which edge would have been the final edge to complete the minimum spanning tree. (2)
- (Total 10 marks)**

Q15.

- (a)
 - (i) State the number of edges in a minimum spanning tree of a network with 11 vertices. (1)
 - (ii) State the number of edges in a minimum spanning tree of a network with n vertices. (1)
- (b) The following network has 11 vertices, $A, B, \dots, K$. The number on each edge represents the distance, in miles, between a pair of vertices.



- (i) Use Prim's algorithm, starting from A , to find a minimum spanning tree for the network.

(5)

- (ii) Find the length of your minimum spanning tree.

(1)

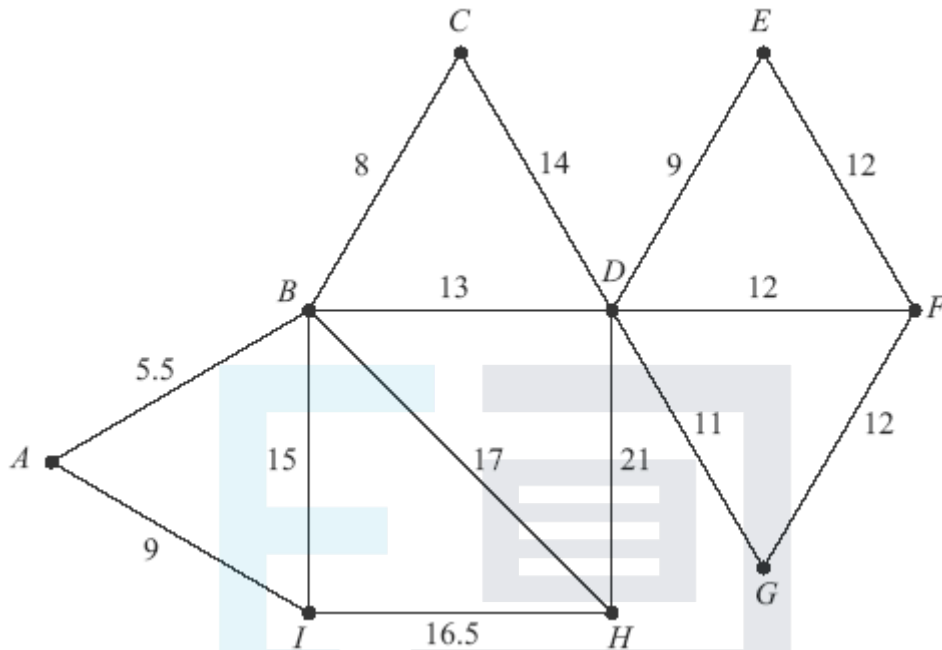
- (iii) Draw your minimum spanning tree.

(2)

(Total 10 marks)

Q16.

The following network shows the lengths, in miles, of roads connecting nine villages.



- (a) Use Prim's algorithm, starting from A , to find a minimum spanning tree for the network.

(5)

- (b) Find the length of your minimum spanning tree.

(1)

- (c) Draw your minimum spanning tree.

(3)

- (d) State the number of other spanning trees that are of the same length as your answer in part (a).

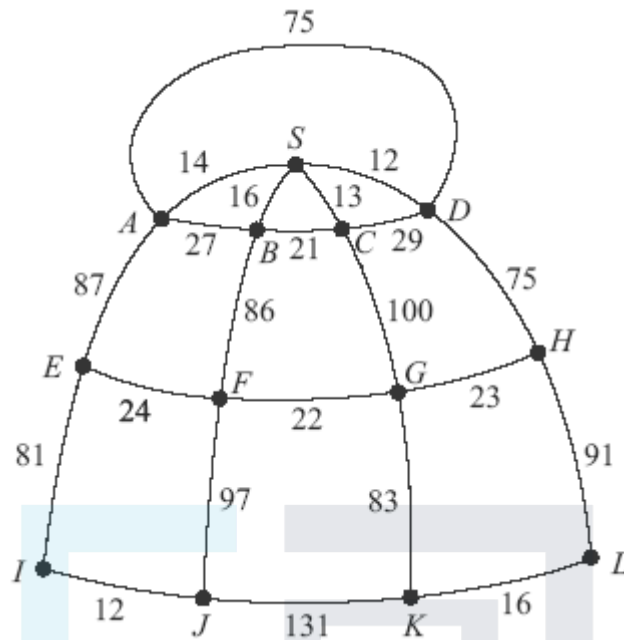
(1)

(Total 10 marks)

Q17.

The diagram shows the various ski-runs at a ski resort. There is a shop at S . The manager of the ski resort intends to install a floodlighting system by placing a floodlight at each of the 12 points $A, B, \dots, L$ and at the shop at S .

The number on each edge represents the distance, in metres, between two points.



Total of all edges = 1135

- (a) The manager wishes to use the minimum amount of cabling, which must be laid along the ski-runs, to connect the 12 points $A, B, \dots, L$ and the shop at S .
- (i) Starting from the shop, and showing your working at each stage, use Prim's algorithm to find the minimum amount of cabling needed to connect the shop and the 12 points. (5)
- (ii) State the length of your minimum spanning tree. (1)
- (iii) Draw your minimum spanning tree. (3)
- (iv) The manager used Kruskal's algorithm to find the same minimum spanning tree. Find the seventh and the eighth edges that the manager added to his spanning tree. (2)
- (b) At the end of each day a snow plough has to drive at least once along each edge shown in the diagram in preparation for the following day's skiing. The snow plough must start and finish at the point L .

Use the Chinese Postman algorithm to find the minimum distance that the snow plough must travel.

(6)
(Total 17 marks)

Q18.

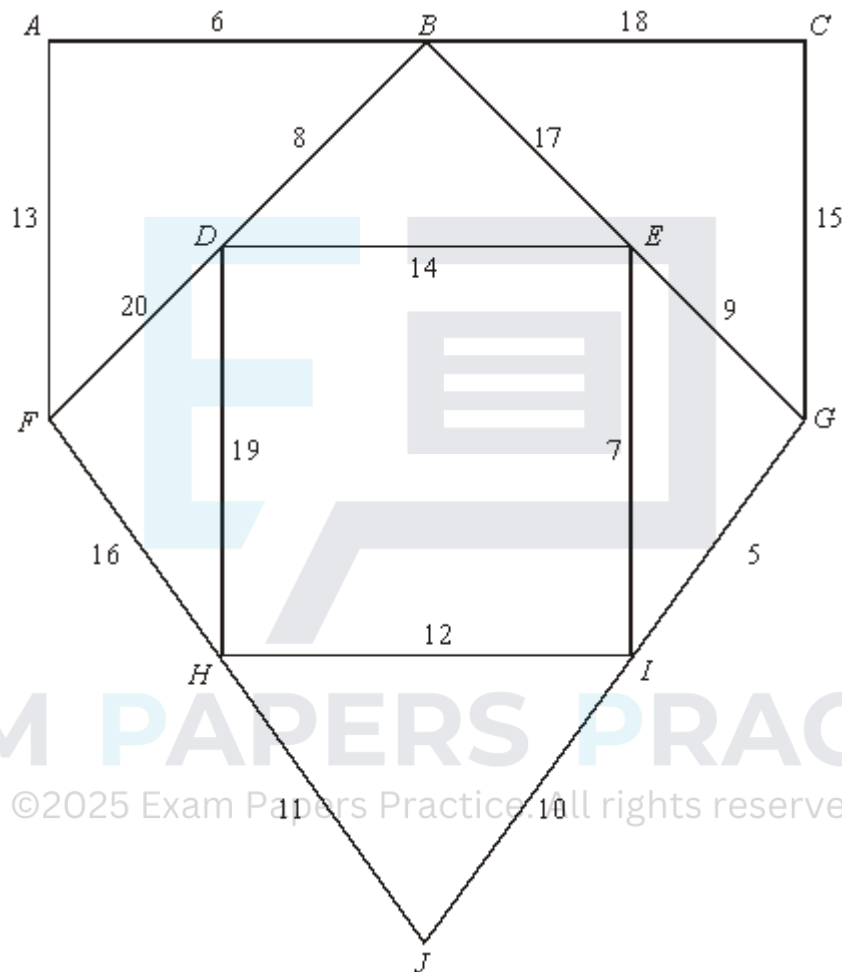
- (a) (i) State the number of edges in a minimum spanning tree of a network with 10 vertices.

(1)

- (ii) State the number of edges in a minimum spanning tree of a network with n vertices.

(1)

- (b) The following network has 10 vertices: $A, B, \dots, J$. The numbers on each edge represent the distances, in miles, between pairs of vertices.



- (i) Use Kruskal's algorithm to find the minimum spanning tree for the network.

(5)

- (ii) State the length of your spanning tree.

(1)

- (iii) Draw your spanning tree.

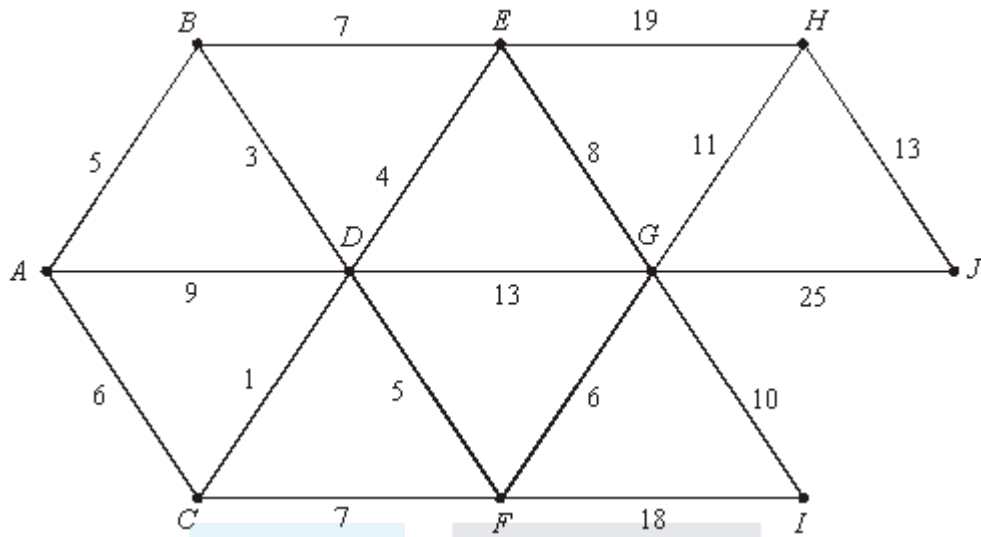
(2)

(Total 10 marks)

Q19.

[Figure 1, is provided for use in **part (b)** of this question.]

The diagram shows a network of roads. The number on each edge is the length, in kilometres, of the road.



- (a) Use Prim's algorithm, starting from A, to find a minimum spanning tree for the network.
- (b) State the length of your minimum spanning tree.

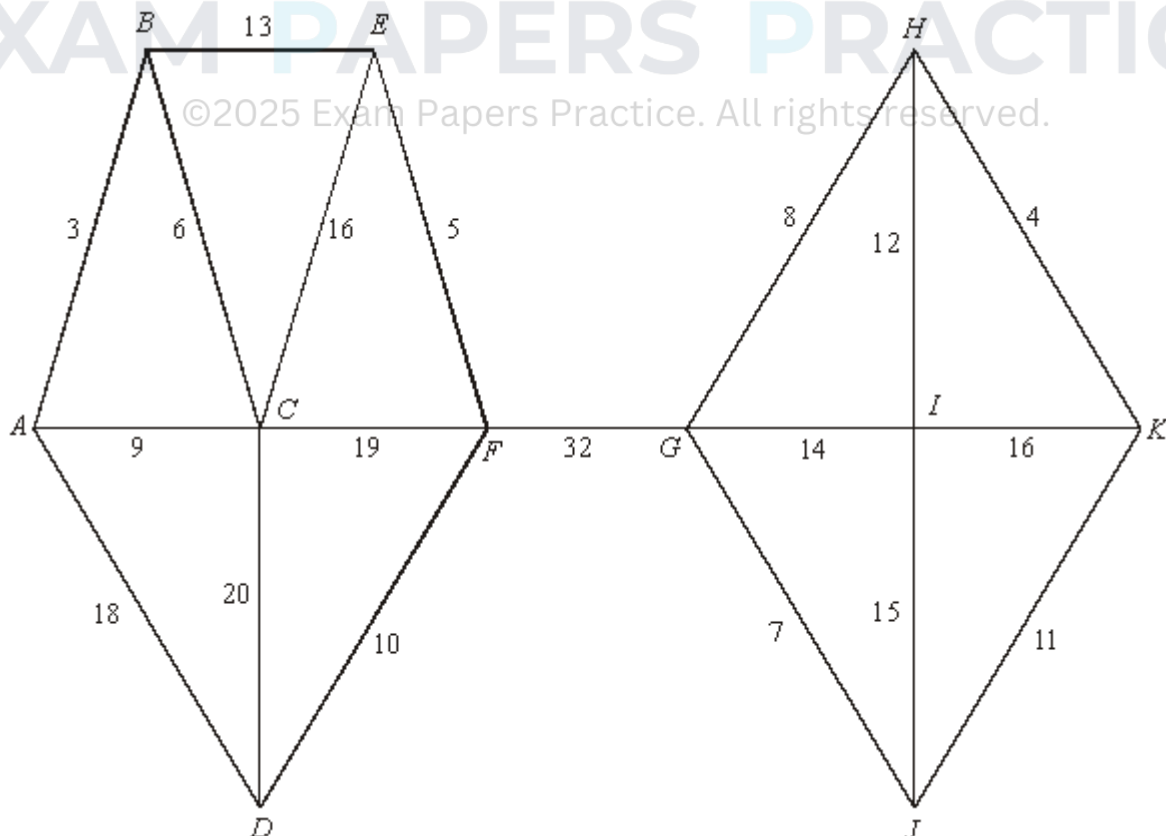
(5)

(1)

(Total 6 marks)

Q20.

The network shows the lengths, in miles, of roads connecting eleven villages.



- (a) Use Prim's algorithm, starting from A, to find the minimum spanning tree for the

network.

(4)

(b) State the length of your minimum spanning tree.

(1)

(c) Draw your minimum spanning tree.

(3)

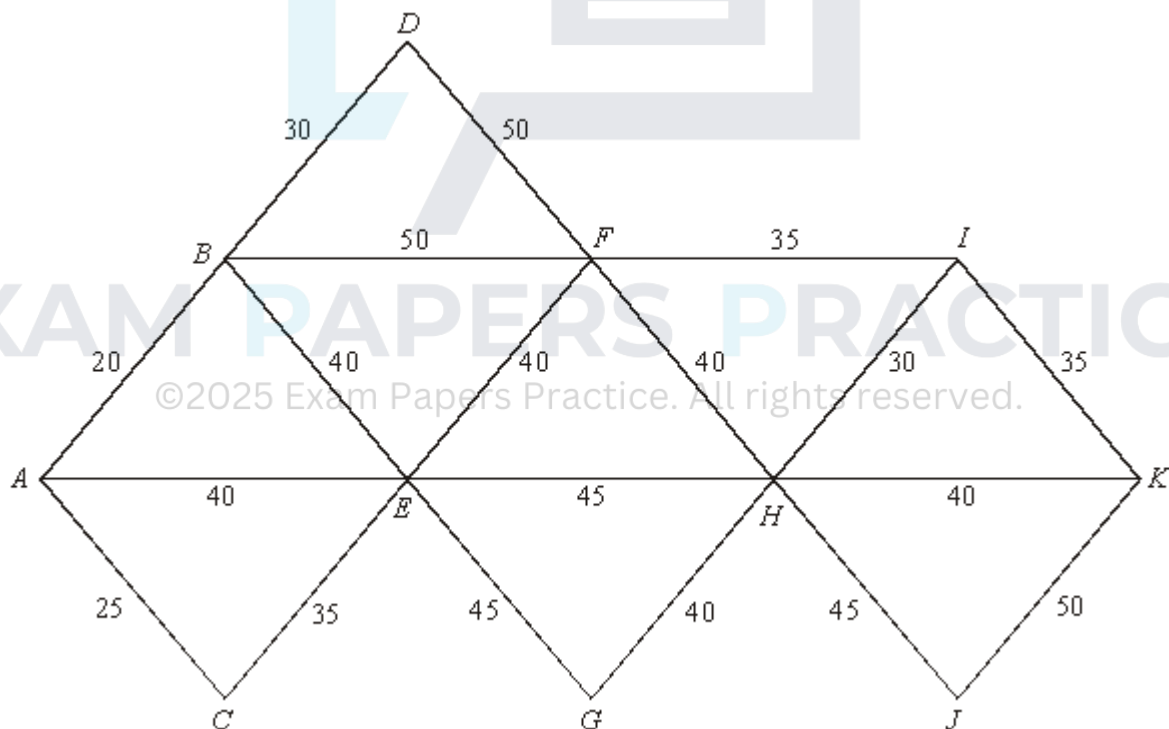
(d) A student used Kruskal's algorithm to find the same minimum spanning tree. Find the seventh and eighth edges that the student added to his spanning tree.

(2)

(Total 10 marks)

Q21.

A theme park has 11 rides, $A, B, \dots, K$. The network shows the distances, in metres, between pairs of rides. The rides are to be connected by cabling so that information can be collated. The manager of the theme park wishes to use the minimum amount of cabling.



(a) Use Prim's algorithm, starting from A , to find the minimum spanning tree for the network.

(5)

(b) State the length of cabling required.

(1)

(c) Draw your minimum spanning tree.

(3)

(d) The manager decides that the edge AE must be included. Find the extra length of

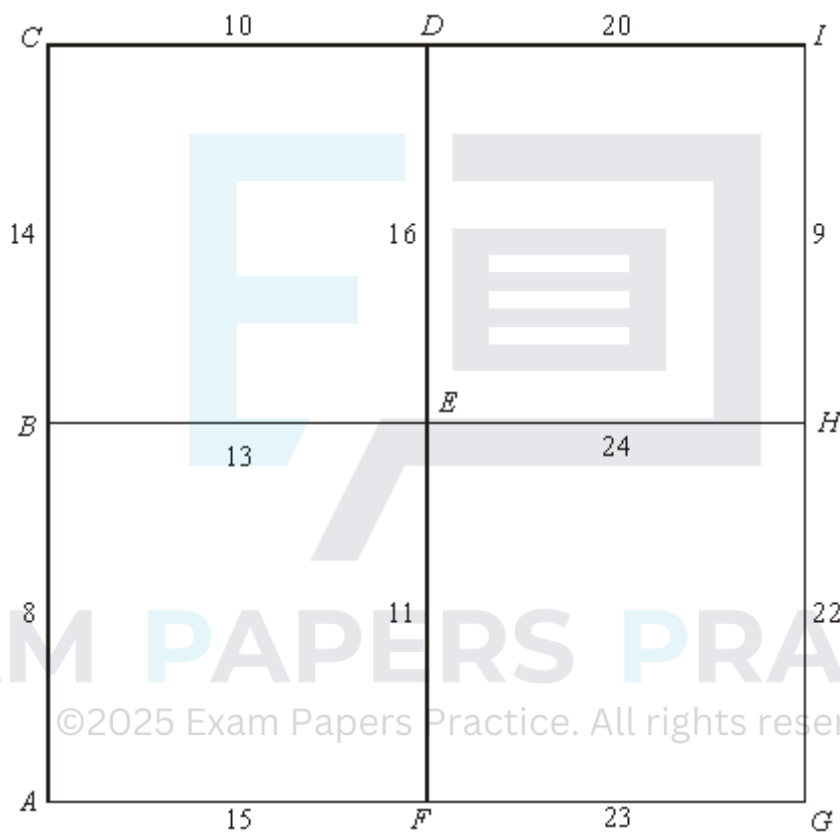
cabling now required to give the smallest spanning tree that includes AE .

(2)

(Total 11 marks)

Q22.

The following network has 9 vertices. The numbers represent the weights of the edges.



- (a) Use Kruskal's algorithm, showing the order in which you select the edges, to find the minimum spanning tree for the network.

(4)

- (b) State the weight of your minimum spanning tree.

(1)

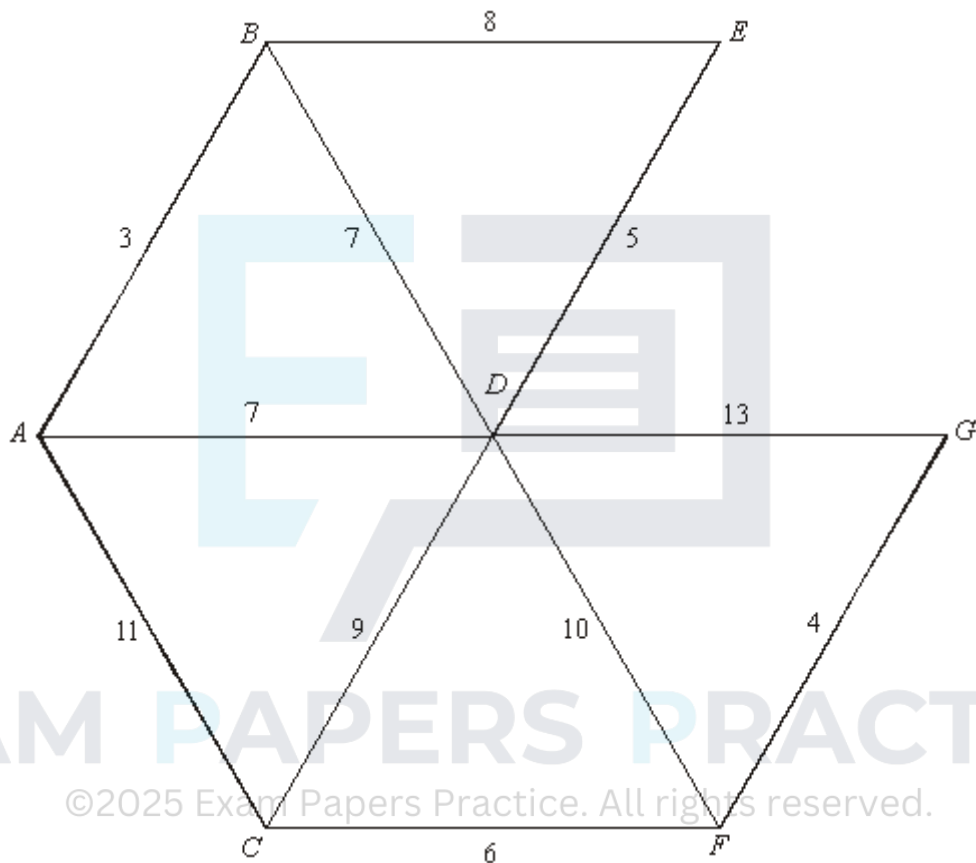
- (c) Draw your minimum spanning tree.

(1)

(Total 6 marks)

Q23.

The following network shows the lengths, in miles, of roads connecting seven villages.



- (a) Use Kruskal's algorithm to find a minimum spanning tree for the network. (4)
- (b) State the length of your minimum spanning tree. (1)
- (c) There are two minimum spanning trees for this network.

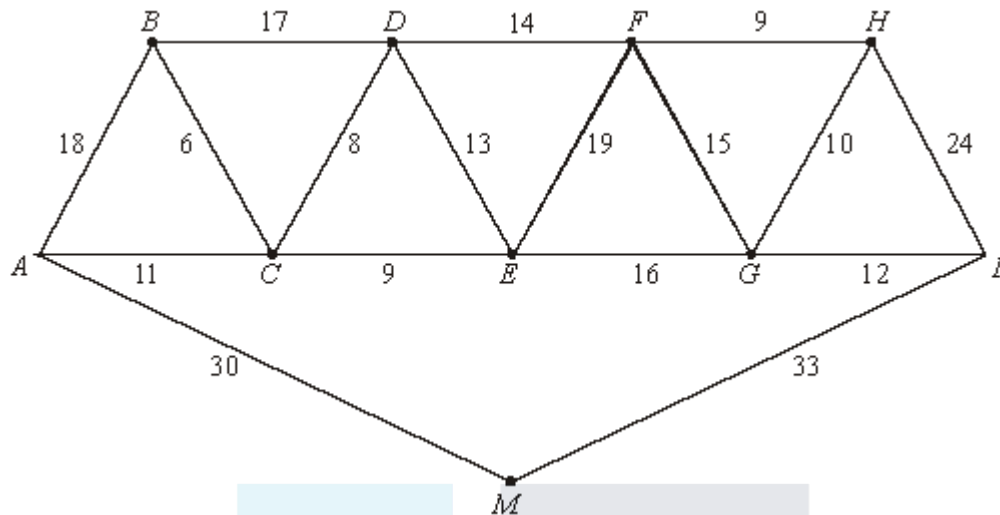
Draw both of these minimum spanning trees.

(2)

(Total 7 marks)

Q24.

Every Christmas Sam puts lights on nine trees, A, B, C, D, E, F, G, H and I , in his garden. The trees have to be connected together either directly or indirectly by cabling and also connected to the mains electricity supply, M , at the house. The cabling has to be laid alongside the garden paths. The diagram shows the lengths of the paths, in metres.



The total length of the paths is 264 metres.

- (a) (i) Use Kruskal's algorithm to find the minimum length of cabling required. State this minimum length.

(5)

- (ii) Draw the minimum spanning tree.

(2)

- (b) The table shows the minimum lengths of paths between A , B , H and I .

	A	B	H	I
A	–	17	42	48
B	17	–	37	43
H	42	37	–	22
I	48	43	22	–

Sam decides to walk along all the paths, starting and finishing at M .

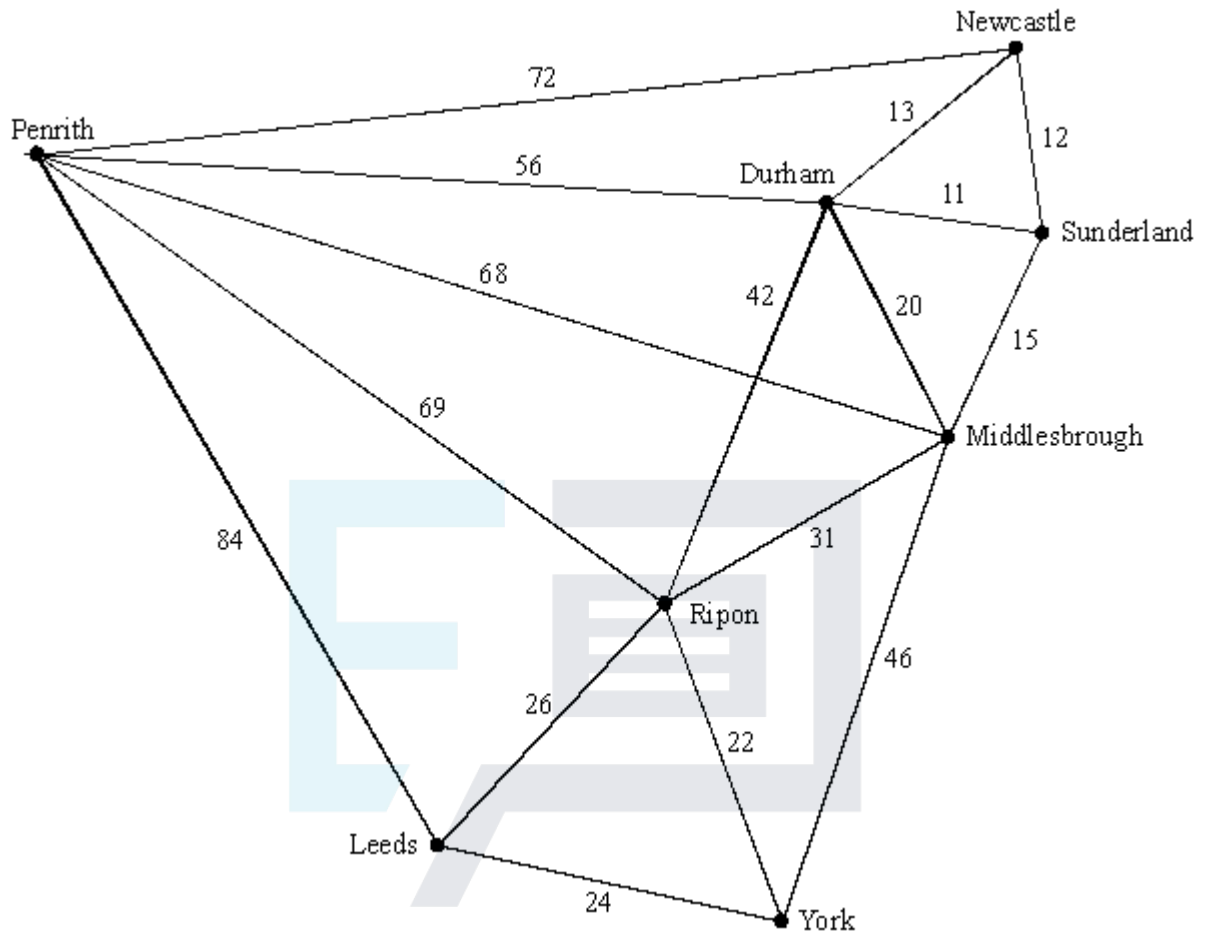
Find the length of an optimal Chinese Postman route for Sam.

(4)

(Total 11 marks)

Q25.

The following diagram shows a network of roads connecting eight towns. The number on each arc represents the distance, in miles, between two towns.



- (a) Starting from Ripon and showing your working at each stage, use Prim's algorithm to find the minimum spanning tree for the eight towns. State the length of your minimum spanning tree.

(5)

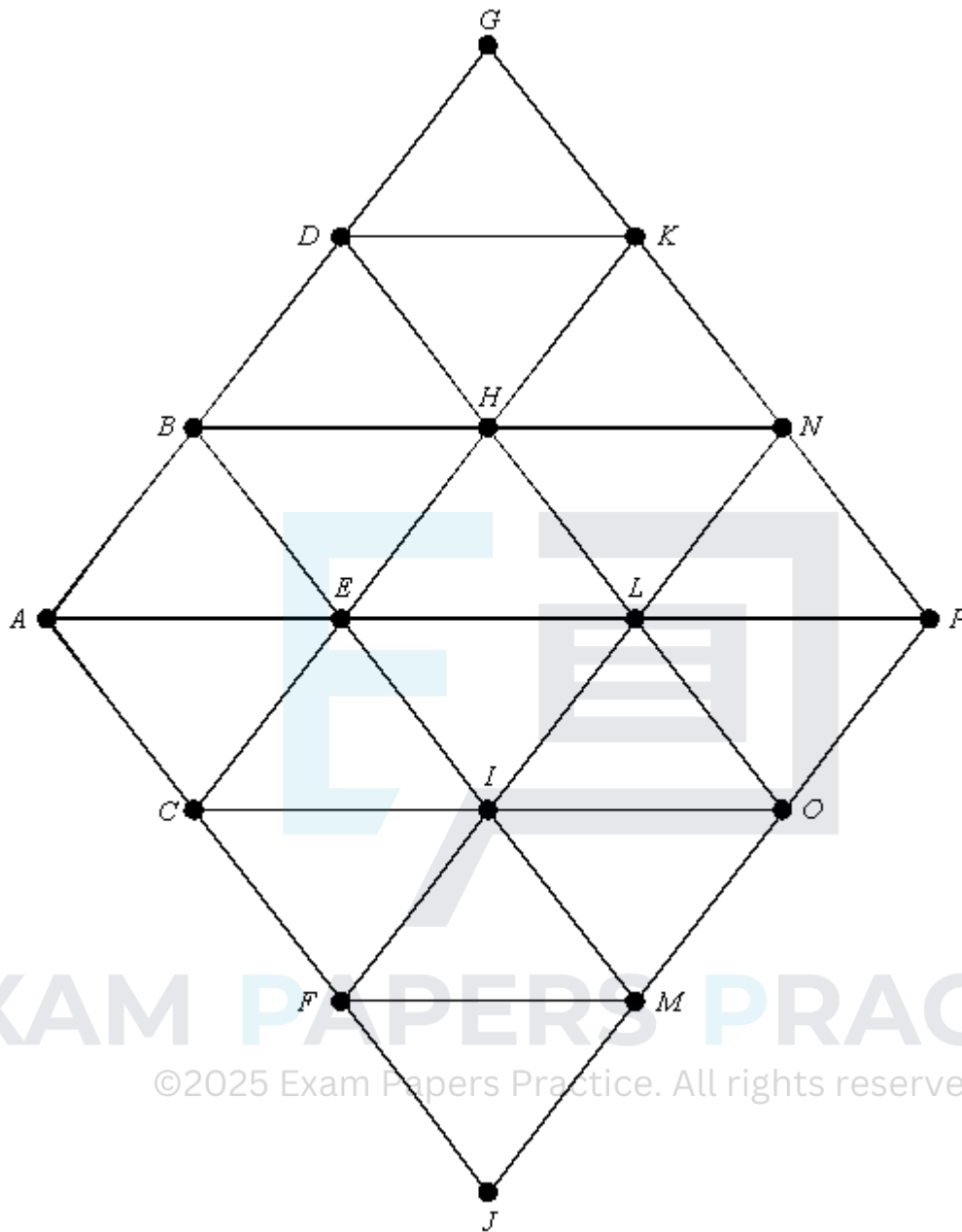
- (b) Draw your minimum spanning tree.

(1)

(Total 6 marks)

Q26.

The following network has 16 vertices.

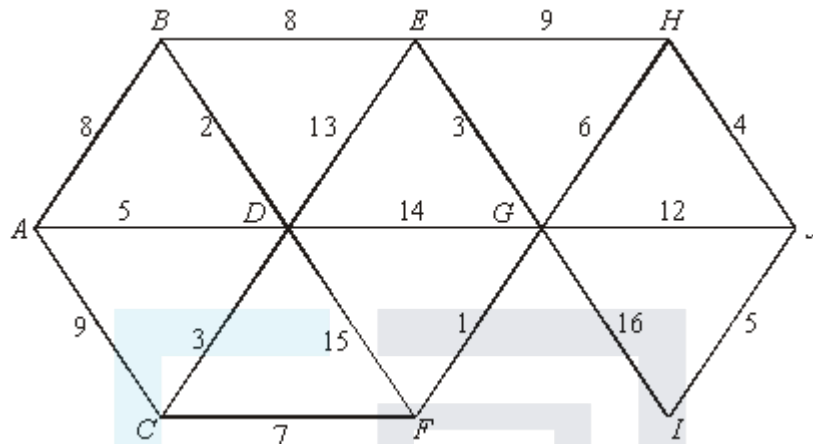


- (a) Given that the length of each edge is 1 unit, find:
- (i) the shortest distance from A to K ; (1)
 - (ii) the length of a minimum spanning tree. (2)
- (b) (i) Find the length of an optimal Chinese Postman route, starting and finishing at A . (4)
- (ii) For such a route, state the edges that would have to be used twice. (1)
- (iii) Given that the edges AE and LP are now removed, find the new length of an optimal Chinese Postman route, starting and finishing at A .

(2)
(Total 10 marks)

Q27.

The following diagram shows the lengths, in miles, of roads connecting ten towns.

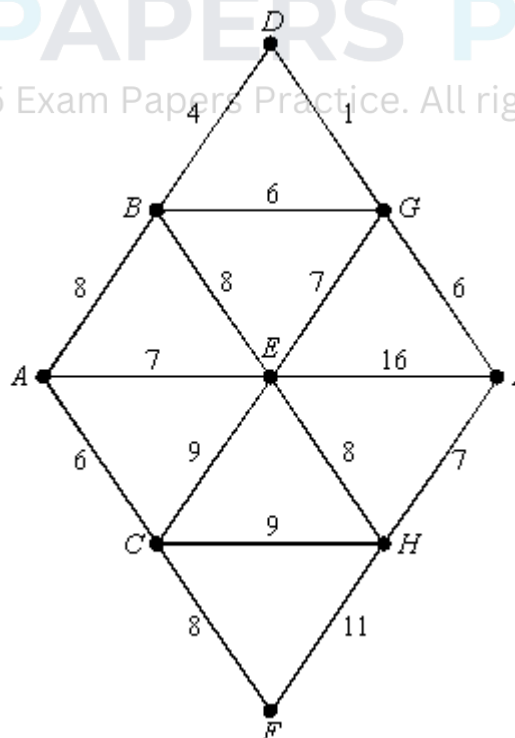


Use Kruskal's algorithm, showing the order in which you select the edges, to find the minimum spanning tree for the network. Draw your minimum spanning tree and state its length.

(Total 6 marks)

Q28.

The following diagram shows the lengths of roads, in miles, connecting nine towns.



- (a) Use Prim's algorithm starting from A, showing your working at each stage, to find the minimum spanning tree for the network. State its length.

(6)

- (b) (i) Find an optimal Chinese Postman route around the network, starting and finishing at A. You may find the shortest distance between any two towns by inspection.

(4)

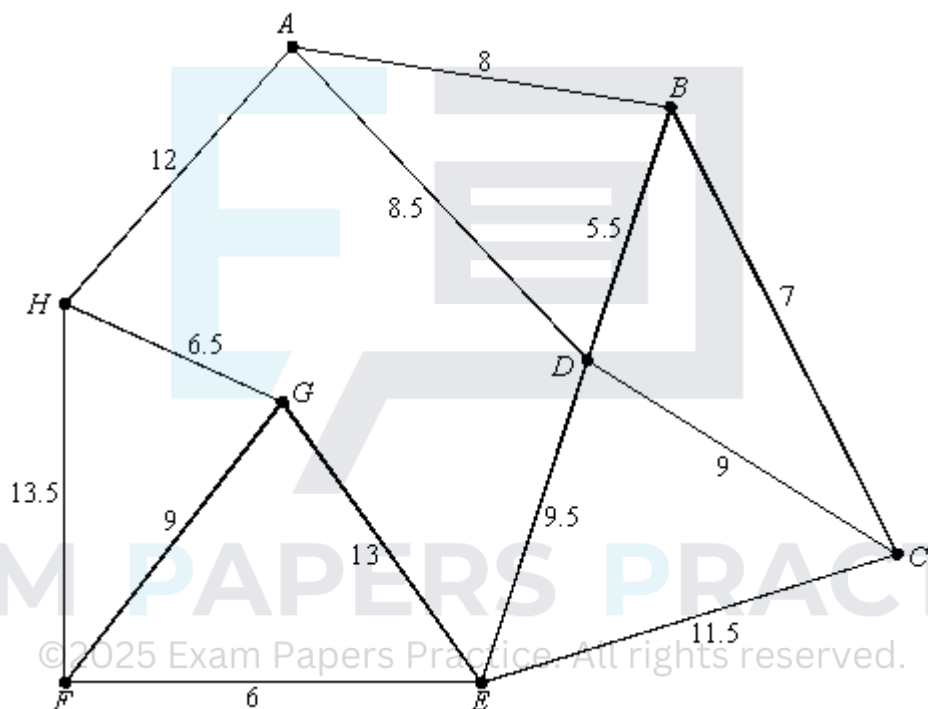
- (ii) State the length of your route.

(1)

(Total 11 marks)

Q29.

The following diagram shows the lengths of roads, in miles, connecting eight towns.



Using Kruskal's algorithm, showing your working at each stage, find the minimum spanning tree for the network. State its length.

(Total 6 marks)

Q30.

- (a) A connected graph has four vertices. State the number of edges in the graph's minimum spanning tree

(1)

- (b) A graph has n vertices. The graph is complete, i.e. each vertex is joined to every other vertex by exactly one edge.

- (i) State the number of edges in the graph's minimum spanning tree

(1)

- (ii) Determine the number of Hamiltonian cycles in the graph.

(2)

- (c) A connected graph has four vertices and has arc lengths of

4, 4.5, 5, 6.5, 7, 8 and 9 units.

The length of its minimum spanning tree is 17 units. Draw a sketch to show a possible graph.

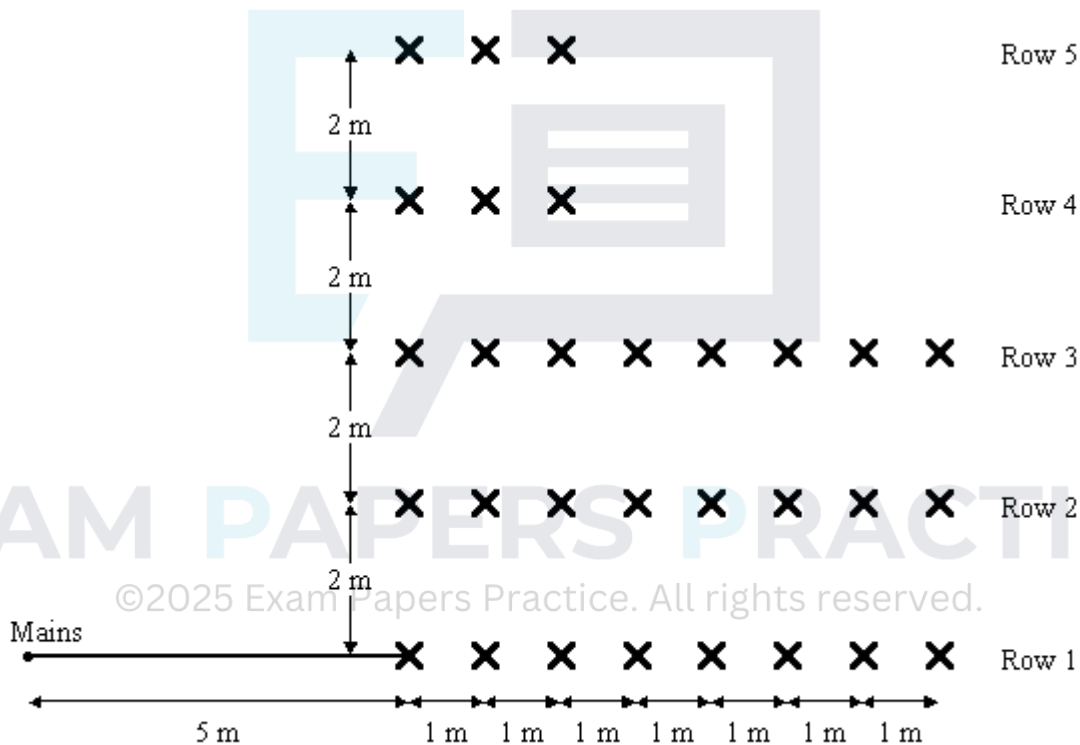
(3)

(Total 7 marks)

Q31.

The government commissions the growing of a certain species of plant to assist in trials of a new drug. The plants must be grown in controlled conditions which necessitate regular watering using an automated sprinkler system. Each plant must have its own sprinkler outlet at its base.

The plants stand 1 metre apart in rows. The rows are 2 metres apart. The mains water supply is 5 metres from the first plant, as shown in the following diagram.



The minimum length of piping is to be used in the sprinkler system.

- Show that the minimum length of piping needed, if 365 plants are to be planted with 50 plants in each of seven rows and 15 plants in an eighth row, is 376 metres. (3)
- Find the minimum length of piping needed if 30 plants are to be planted. (1)
- Find, in its simplest form, an expression for the minimum length of piping needed to plant m rows with each row containing exactly n plants. (3)
- It is required to plant x plants, where x is a multiple of 60, in rows each containing 60 plants. Show that the minimum length of piping needed is $ax + b$, where a and b are numbers to be determined.

(2)
(Total 9 marks)

Q32.

The distance chart below gives the distances, in miles, between six towns in northern England.

Bradford					
10	Halifax				
20	30	Harrogate			
12	8	32	Huddersfield		
13	22	19	24	Leeds	
15	22	32	14	12	Wakefield

- (a) Using Kruskal's algorithm and showing your working at each stage, find a minimum spanning tree for these six towns.

(4)

- (b) State the length of your minimum spanning tree.

(1)

(Total 5 marks)

Q33.

A connected graph G has five vertices. The lengths of the edges are 7, 7, 7, 9, 11, 12, 13, 15 and 16 units, respectively.

- (a) A graph is said to be *fully connected* if every vertex is joined at least once to every other vertex. Explain why graph G cannot be fully connected.

(1)

- (b) A connected graph G is to be drawn with 5 vertices with lengths of edges as given above

- (i) Calculate the least possible length of a minimum spanning tree of graph G .

(1)

- (ii) Explain why your answer to part (b)(i) might not apply to graph G .

(1)

- (iii) Sketch an example of graph G which has a minimum spanning tree of length 34 units.

(3)

(Total 6 marks)

Q34.

The distances, in miles, between six towns are given by the following matrix.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>
<i>A</i>	–	3	5	10	13	19
<i>B</i>	3	–	4	7	13	23
<i>C</i>	5	4	–	7	10	22
<i>D</i>	10	7	7	–	17	18
<i>E</i>	13	13	10	17	–	9
<i>F</i>	19	23	22	18	9	–

- (a) Using Prim's algorithm and showing your working at each stage, find a minimum spanning tree for these six towns. (5)
- (b) State the length of your minimum spanning tree. (1)
- (c) When information is provided in matrix form, explain why Prim's algorithm, in preference to Kruskal's algorithm, is normally used to find a minimum spanning tree. (1)
- (Total 7 marks)**

Q35.

The following matrix shows the distances, in miles, between six towns.

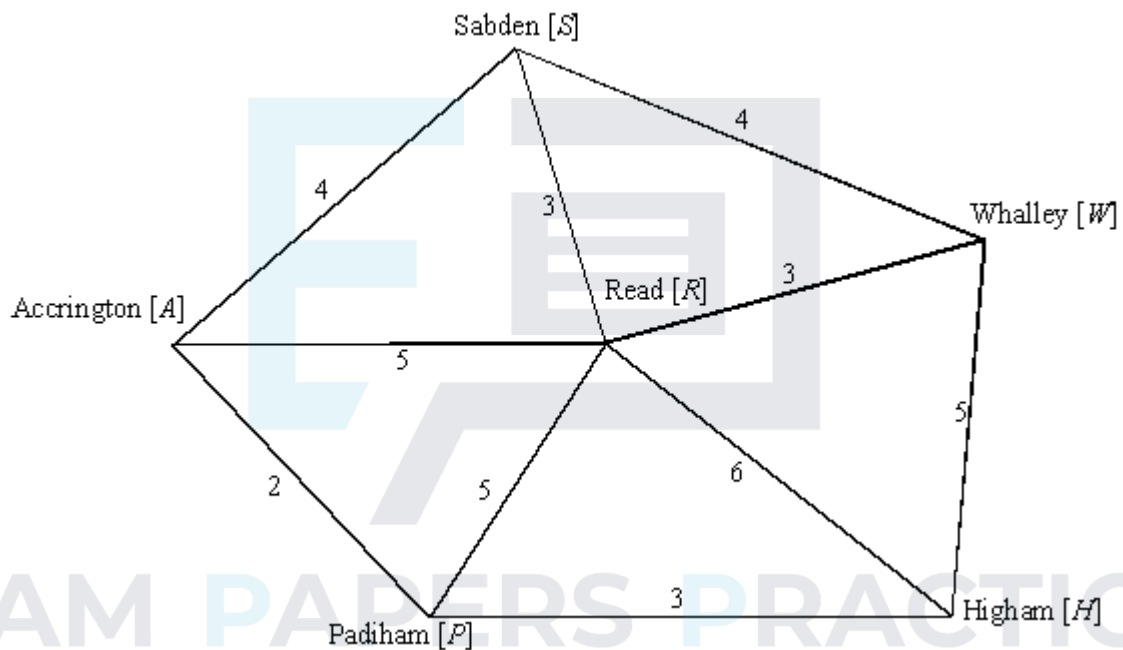
	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>
<i>A</i>	—	12	20	14	15	25
<i>B</i>	12	—	23	7	13	8
<i>C</i>	20	23	—	16	16	12
<i>D</i>	14	7	16	—	20	9
<i>E</i>	15	13	16	20	—	15
<i>F</i>	25	8	12	9	15	—

- (a) Using Kruskal's algorithm and showing your working at each stage, find the minimum spanning tree for these six towns. (4)
- (b) State the length of your minimum spanning tree (1)

(Total 5 marks)

Q36.

- (a) For a connected graph with n vertices, state the number of edges in a minimum spanning tree. (1)
- (b) A cable company has laid cables to Accrington and is now considering extending the network to neighbouring villages. The distances, in miles, between the villages are shown.



Using Prim's algorithm, showing a sketch at each stage, obtain the minimum spanning tree for the cable company.

- (5)
- (c) State the length of your minimum spanning tree. (1)

(Total 7 marks)

Q37.

A group of seven friends, P , Q , R , S , T , U and V , keep in touch by phone. The minimum cost, in pence, of a phone call between any pair of them is shown in the following table.

	P	Q	R	S	T	U	V
P	–	30	50	45	30	35	60
Q	30	–	40	40	80	70	55
R	50	40	–	45	80	70	70

S	45	40	45	–	35	60	60
T	30	80	80	35	–	30	60
U	35	70	70	60	30	–	50
V	60	55	70	60	60	50	–

Person *P* wishes to pass on a piece of news to all the other friends, either by a direct phone call or by the message being passed on from friend to friend.

- (a) By applying Prim's algorithm to the matrix in the figure above, find the minimum cost of notifying all the friends of the news.

(5)

- (b) Draw a graph representing the minimum connector which you found in part (a).

(2)

- (c) Person *Q* is on holiday and so a message can be left on her answerphone but she cannot be used to pass the message on. In these new circumstances, what is the minimum cost of notifying all the friends of the news?

(2)

(Total 9 marks)

Q38.

The walking times, in minutes, between six points *A* – *F* are given in the following table:

	A	B	C	D	E	F
A	–	20	30	20	15	10
B	20	–	35	30	25	25
C	30	35	–	30	30	25
D	20	30	30	–	15	25
E	15	25	30	15	–	10
F	10	25	25	25	10	–

- (a) Use Prim's algorithm on the table above, starting at *A*, to find a minimum connector of the six points.

(5)

- (b) Draw a tree representing the minimum connector which you found in part (a).

(2)

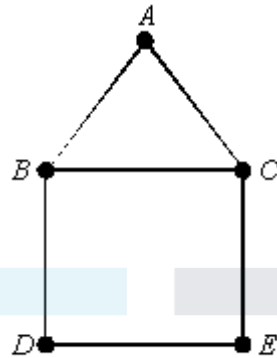
- (c) Due to poor weather, only the footpaths in the minimum connector are open. Use your tree from part (b) to show that it is still possible to walk from any one of the six

points to any other in less than 1 hour.

(3)
(Total 10 marks)

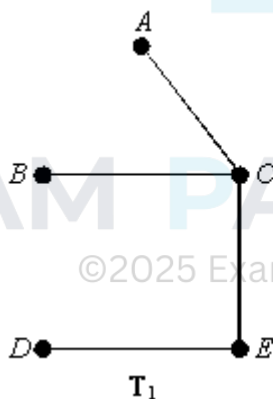
Q39.

The graph **G** is illustrated:



Graph G

Two of **G**'s spanning trees (or minimum connectors, with all the edges of equal weight) **T**₁, and **T**₂ are shown below. In each of these trees the degree of the vertex **A** equals the degree of the vertex **D**.



- (a) Draw **three** other spanning trees of **G** in which the degree of vertex **A** equals the degree of vertex **D**.

(6)

- (b) For each of the trees which you found in part (a), state whether it is isomorphic to **T**₁, or to **T**₂ or to neither.

(3)
(Total 9 marks)

Q40.

The following table shows the lengths, in miles, of pipe-lines between six water-pumping stations **A** – **F**.

A	B	C	D	E	F
----------	----------	----------	----------	----------	----------

A	—	23	17	19	18	22
B	23	—	20	24	23	21
C	17	20	—	24	19	23
D	19	24	24	—	20	24
E	18	23	19	20	—	24
F	22	21	23	24	24	—

- (a) The authorities decide to disconnect some of the pipe-lines. Use Prim's algorithm on above, starting at *A*, to find the minimum total length of pipe-lines which need to be kept open in order for all the pumping stations to remain linked together.
- (b) The pipe-line from *B* to *C* is found to be faulty. So when choosing the pipe-lines to keep open the authorities must not include *BC*. By how many miles does this increase the total length of pipe-lines which need to be kept open?

(5)

(3)

(Total 8 marks)

Q41.

The table shows the distances, in miles, of any direct routes between the seven towns *A*, *B*, *C*, *D*, *E*, *F* and *G*:

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>
<i>A</i>	—	5	15	—	—	—	7
<i>B</i>	5	—	8	—	—	9	15
<i>C</i>	15	8	—	7	13	—	6
<i>D</i>	—	—	7	—	5	6	10
<i>E</i>	—	—	13	5	—	5	—
<i>F</i>	—	9	—	6	5	—	6
<i>G</i>	7	15	6	10	—	6	—

- (a) Use Prim's algorithm to find a minimum connector of the seven towns, and state its length.
- (b) The local authority decides to charge tolls on some of these roads, but it must still be possible to travel between any two towns on toll-free roads. What is the

(5)

maximum length of roads on which it can charge tolls?

(2)

- (c) Opposition from local residents forces the authority to keep the road from B to G free of tolls (and to leave enough other toll-free roads so that it is still possible to travel between any two towns on toll-free roads). With this extra restriction, what is the maximum length of roads on which the local authority can charge tolls?

(3)

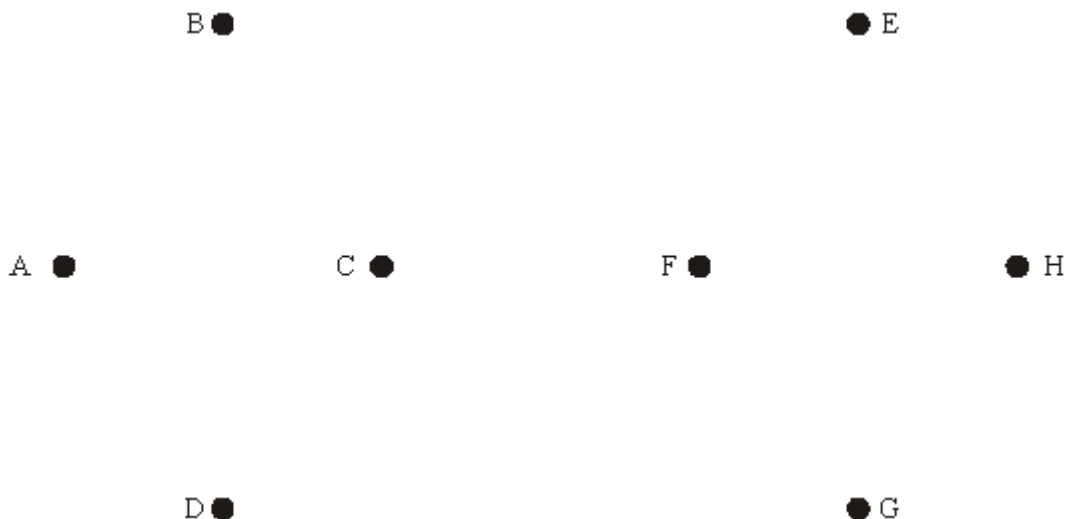
(Total 10 marks)

Q42.

An international organisation has offices A, B, C, D, E, F, G and H . The table shows the cost, in pounds, of transmitting a piece of information from one of the offices to another along all existing direct links.

From \ To	A	B	C	D	E	F	G	H
A	—	12	8	9	—	—	—	—
B	12	—	6	—	10	—	—	—
C	8	6	—	10	—	10	—	—
D	9	—	10	—	—	—	12	—
E	—	10	—	—	—	11	—	20
F	—	—	10	—	11	—	18	18
G	—	—	—	12	—	18	—	14
H	—	—	—	—	20	18	14	—

- (a) Construct a network using the vertices A to H below to represent the information in the table.



(2)

- (b) A piece of information has to be passed from office A to all the other offices, either

directly or by being passed on from office to office. Use Kruskal's algorithm to find the minimum total cost of passing the information to all the offices.

(5)

- (c) Office *C* joins a communications discount scheme in which the cost of passing information from *C* is halved (but it does not affect the cost of *C*'s incoming information).

(i) Describe how to adapt the above table to show the revised costs.

(1)

- (ii) Adapt your answer to part (b) in order to calculate the minimum cost of passing a piece of information from office *A* to all the other offices with these revised costs.

(4)

(Total 12 marks)

Q43.

Some of the eight towns *A* to *H* are directly linked by roads. The table shows the distances along these roads in miles.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>
<i>A</i>	—	10	9	5	7	6	7	—
<i>B</i>	10	—	7	8	—	—	8	9
<i>C</i>	9	7	—	—	7	8	—	8
<i>D</i>	5	8	—	—	—	9	9	—
<i>E</i>	7	—	7	—	—	8	9	7
<i>F</i>	6	—	8	9	8	—	6	9
<i>G</i>	7	8	—	9	9	6	—	—
<i>H</i>	—	9	8	—	7	9	—	—

- (a) In the winter the local authority grits some of these roads. They wish to grit the minimum total length of roads possible in order to be able to travel between any two of the towns on gritted roads. Use Prim's algorithm starting at *A* to find which roads they should grit, and state the minimum total length of roads which need to be gritted.

(6)

- (b) Illustrate in a graph the minimum connector which you found in (a). How far is it from *A* to *B* in your graph?

(3)

- (c) The ambulance station is at *A* and next year the local authority decides that it would

like to grit the minimum length of roads possible so that the ambulance can reach any of the other towns on gritted roads by travelling less than 15 miles. Show how to adapt your graph in (b) to find the roads which the local authority should grit next year.

(3)

(Total 12 marks)



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